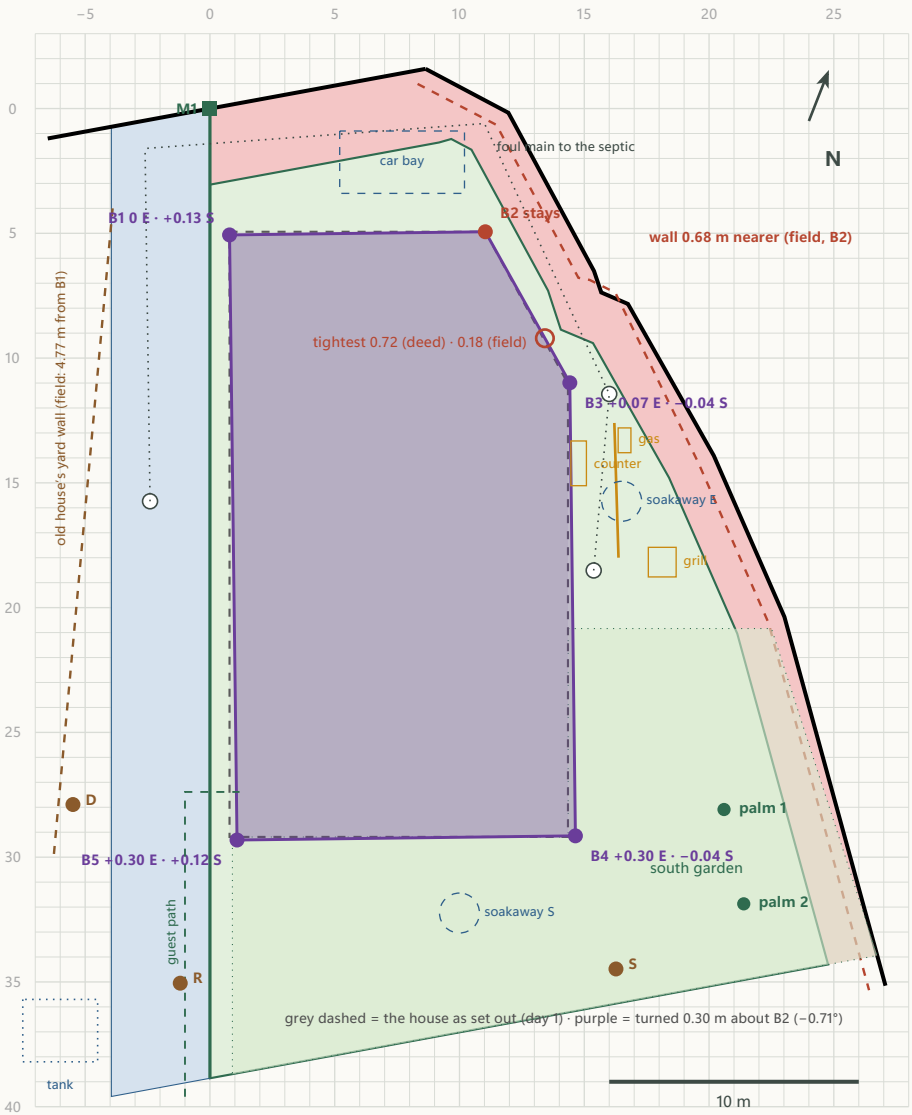


Taif / Palm House

Rotation about B2 · study (owner's idea, 5 Nov 2026) · STUDY ONLY, nothing moved

Frame: origin M1 · y along the M line southwards · x east. B2 (the tightest corner) stays; the house turns so that its south end moves EAST, away from the driveway: B5 east by 0.10 … 0.40 m. The house as set out on day 1 (B1 5.00 m from M1) is the base. Page 2: the comparison of the options · page 3: the recommended option's set-out table.

Recommendation: B5 east by 0.30 m, after the notch check

The house turns 0.71° about B2. Against the deed setback lines the tightest margin goes from 0.79 to **0.72 m** (the cut wall at the lane's notch), above the 0.50 m spare; B2 unchanged; no conflict with the palms, the garden, the guest path, the outdoor kitchen, the gas cabinet, the tanks, the sewer, the car bay. The route beside the driveway keeps 4.77 m at B1 and widens to 7.29 m at B5. **Cost:** B1 moves 0.13 m south, so the owner's rule "B1 = 5.00 m from M1" becomes 5.13 m; the garden loses 2.65 m² beside the east wall. **Condition:** if the setback is counted from the wall as measured at B2, the cut wall is tight already as set out (B2 0.40 m; the notch 0.24 m if the wall runs 0.68 m inside the deed line all along); the rotation costs 0.06 m there. Do the notch check (sheet row 18) first; the decision stands: no reduction of the cut wall until the official survey.

1 · The field values of day 1 against the model

Item	Field	Model	Diff.	Likely cause · what the study does
B2 square to the boundary wall's inner face (2.60 incl. the wall)	**2.40**	3.08 to the deed line (margin 1.08 to the setback)	−0.68	the wall stands inside the deed line (the deed line is the plot's limit, the wall is built inside it; the v2 frame's rms 0.27 m; B2 itself was set +0.07 from M1). **Calibrated:** the wall taken 0.68 m nearer everywhere (conservative) and the setback counted 2.0 m from its inner face → B2's margin 0.40 m instead of 1.08.
B2 is the nearest point of the house to the wall (owner)	yes	the cut wall at (13.1, 8.8) is 2.92 from the deed line, B2 3.08	0.16	the model's nearest point is 2.6 m along the cut wall from B2, at the lane's notch, 0.16 m nearer than B2. If B2 really is the nearest, the wall at the notch is ≥ 2.40 m away (the deed line says 2.92): the field check of the notch (sheet row 18) decides.
B1 to the old house's yard wall across the driveway (the route's narrowest point, widening south)	**4.77**	4.74 to the driveway's west edge (v2); 6.94 to the deed-sketch old house	+0.03	the tape (3 cm): the yard wall IS the driveway's west edge there; the deed-sketch old house is ≈ 2 m too far west at the north (digitised ± 1–3 m; W4 on the field sheet). **Calibrated:** the yard wall drawn 4.77 m west of B1, running away southwards like the sketch's east face (0.91 m per 10 m), as the owner saw.
Decision (owner, day 1)	—	—	—	**No reduction of the cut wall until the official survey.** The rotation needs none: B2 does not move.

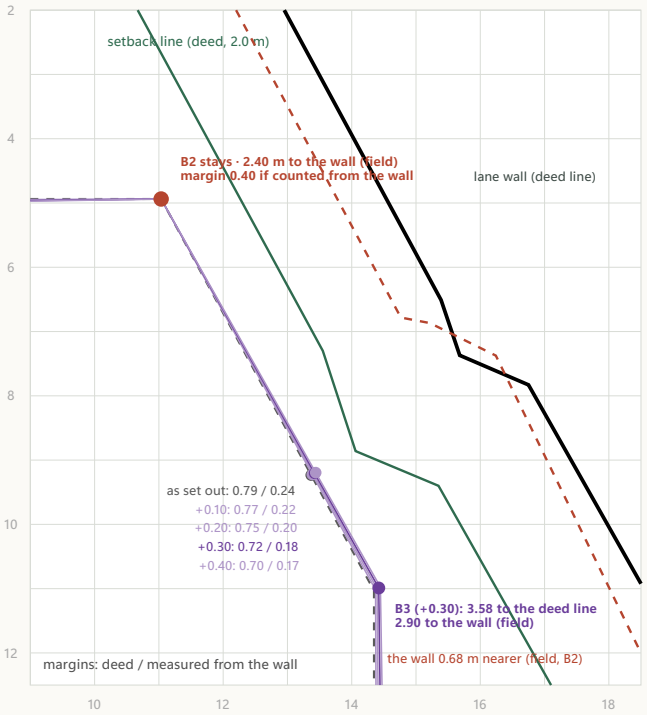
The plan: grey dashed = the house as set out (day 1); purple = turned 0.30 m about B2; the red dashed line = the lane wall 0.68 m nearer than the deed line (the field value at B2, taken along the whole wall); the brown dashed line = the old house's yard wall from the field value at B1. Every old-building position is approximate, to be measured.

The options · B5 east by 0.10 … 0.40 m, B2 fixed

every wall sampled at 5 cm, not only the corners · deed = the setback lines of the deed · "from the wall" = the setback counted 2.0 m from the wall as measured at B2 (0.68 m nearer, conservative)

B5 east by (m)	**as set out**	**0.10**	**0.20**	**0.30** (recommended)	**0.40**
The house turns about B2 by (°)	0.00	0.24	0.47	0.71	0.94
B1 moves south (m) · M1 → B1 (was 5.00, the owner's rule of 4 Nov 2026)	0.00 · **5.00**	0.04 · **5.04**	0.08 · **5.08**	0.13 · **5.13**	0.17 · **5.17**
B3 moves east / north (m)	0.00 / 0.00	0.03 / 0.01	0.05 / 0.03	0.07 / 0.04	0.10 / 0.06
B4 moves east / north (m)	0.00 / 0.00	0.10 / 0.01	0.20 / 0.03	0.30 / 0.04	0.40 / 0.06
B5 moves east / south (m)	0.00 / 0.00	0.10 / 0.04	0.20 / 0.08	0.30 / 0.12	0.40 / 0.17
Tightest margin to the deed setback (on the cut wall at the lane's notch; ≥ 0.50 wanted; B2 unchanged)	**0.79**	**0.77**	**0.75**	**0.72**	**0.70**
The same with the setback counted from the wall (0.68 m nearer everywhere: conservative)	**0.24**	**0.22**	**0.20**	**0.18**	**0.17**
B2's margin: deed / from the wall (unchanged in every option)	1.08 / 0.40	1.08 / 0.40	1.08 / 0.40	1.08 / 0.40	1.08 / 0.40
B3 to the deed line / to the wall as measured · B3's margin to the deed setback	3.67 / 2.99 · 1.68	3.64 / 2.96 · 1.65	3.61 / 2.93 · 1.62	3.58 / 2.90 · 1.59	3.55 / 2.87 · 1.56
The cut wall's nearest point to the deed line (between B2 and B3)	2.92	2.90	2.88	2.87	2.85
North strip at B1: the north face to the north setback line	2.00	2.04	2.08	2.12	2.16
The car bay's south edge to the north wall at the bay's west end (≥ 0.4) · the swept path's car side to the wall (the turn into the bay)	1.54 · 1.89	1.56 · 1.91	1.59 · 1.94	1.61 · 1.96	1.63 · 1.98
West wall to the driveway's west edge at y 10 / 15 / 20 / 25 / at B5 (at B1 4.74, unchanged)	4.74 / 4.74 / 4.74 / 4.74 / 4.74	4.76 / 4.78 / 4.80 / 4.82 / 4.84	4.78 / 4.82 / 4.86 / 4.90 / 4.94	4.80 / 4.86 / 4.92 / 4.98 / 5.04	4.82 / 4.90 / 4.98 / 5.06 / 5.14
West wall to the old house's yard wall (calibrated) at y 10 / 15 / 20 / 25 / at B5 (at B1 4.77, unchanged)	5.23 / 5.68 / 6.14 / 6.59 / 6.97	5.25 / 5.73 / 6.20 / 6.68 / 7.08	5.27 / 5.77 / 6.26 / 6.76 / 7.18	5.29 / 5.81 / 6.32 / 6.84 / 7.29	5.31 / 5.85 / 6.39 / 6.92 / 7.39
West wall to the guest path's line at y 10 / 15 / 20 / 25	1.79 / 1.79 / 1.79 / 1.79	1.81 / 1.83 / 1.85 / 1.87	1.83 / 1.87 / 1.91 / 1.95	1.85 / 1.91 / 1.97 / 2.03	1.87 / 1.95 / 2.03 / 2.11
Palm 1 / palm 2 to the nearest wall (trunk face)	5.99 / 7.27	5.89 / 7.18	5.80 / 7.10	5.70 / 7.01	5.61 / 6.92
South garden: ground lost under the turned house (m², beside the east wall and at the south-west) · B5 to the garden's low wall	0.00 · 0.19	0.87 · 0.28	1.77 · 0.39	2.65 · 0.49	3.53 · 0.59
Guest door moves east / south · the guest path's last leg (from its line to the door)	0 / 0 · 1.91	+0.09 / +0.04 · 2.00	+0.18 / +0.08 · 2.09	+0.28 / +0.12 · 2.19	+0.37 / +0.16 · 2.28
Outdoor kitchen, yard screen, gas cabinet, grill: turn with the house (their spacings unchanged) · the gas cabinet to the deed line / to the wall as measured (≥ 1.5)	2.47 / **1.79**	2.43 / **1.74**	2.38 / **1.70**	2.34 / **1.66**	2.30 / **1.62**
The existing tank to B5 (≥ 5, assumed rule)	**8.39**	**8.42**	**8.45**	**8.48**	**8.52**
Sewer: the yard chamber (turns with the house) → the east strip's chamber (fixed) · the west drain's run from the wall to its chamber	7.17 · 3.31	7.15 · 3.35	7.12 · 3.40	7.10 · 3.44	7.08 · 3.49
Soakaway E to the east wall · soakaway S to the south wall	2.15 · 3.05	2.11 · 3.05	2.06 · 3.04	2.02 · 3.04	1.98 · 3.03
Sides and diagonals	unchanged	unchanged	unchanged	unchanged	unchanged
Conflicts	**none**	**none**	**none**	**none**	**none**

Reading the table. Every option keeps every rule of the services plan and the set-out sheet against the deed lines; the cost grows with the angle: B1 slides south along the M line (the owner's 5.00 m rule), the notch loses 0.02 m per 0.10 m of rotation, the garden loses ground beside the east wall, the east soakaway comes nearer the wall (already below the services plan's assumed 5 m rule as set out: for the engineer, not caused by the rotation). The gain is at the south-west: the route beside the driveway, where the water truck stands and the cars turn into the covered parking, widens by the option; the north strip at B1 deepens by the B1 move. **With the setback counted from the wall as measured:** B2's 0.40 m does not change with any option, and the notch is tight already as set out if the wall runs 0.68 m inside the deed line all along; the owner saw B2 as the nearest point, which says it does not. The notch check (sheet row 18: from the tightest point square to the lane wall, expected 2.92 m by the deed line, at least 2.40 m if B2 is the nearest) settles it before any block moves.



The cut wall enlarged. Grey dashed: as set out. Purple: the options (the recommended one bold). Each dot = the tightest point of that option to the deed setback, labelled with its margin to the deed setback / to a setback counted 2.0 m from the wall as measured at B2 (0.68 m nearer than the deed line, taken the same along the whole wall: conservative; the owner saw B2 as the nearest point, so the wall at the notch may well be further off: sheet row 18 decides).

Recommendation · B5 east by 0.30 m

the owner's preference, checked

Safe against the deed setback lines: the tightest margin falls from 0.79 to **0.72 m** on the cut wall at the notch (a cost of 0.06 m), still above the 0.50 m spare; B3 3.58 m from the deed line; B2 unchanged. Every corner inside the envelope; the palms 5.70 / 7.01 m; the gas cabinet 1.66 m from the wall as measured; the tank 8.48 m from B5; the sewer and the water pipe unchanged in practice (the yard chamber's run to the fixed east chamber 7.17 → 7.10 m; the falls stay); the car bay keeps 1.61 m off the north wall and the swept path clears it by 1.96 m. **Gains:** the route beside the driveway widens from 4.77 m at B1 (unchanged: the narrowest point stays) to 7.29 m at B5 against the yard wall and 5.04 m against the driveway's edge, where the water truck stands and the cars turn into the covered parking; the north strip at B1 deepens by 0.13 m.

What it costs, to decide with open eyes: (1) **the owner's rule "B1 exactly 5.00 m from M1" becomes 5.13 m** (B1 moves 0.13 m south along the M line); (2) the north wall is no longer square to the M line (0.71°: 0.13 m over its 10.25 m, invisible to the eye) and the west wall no longer parallel to the driveway (0.30 m over 24.25 m); (3) the south garden loses 2.65 m² under the turned house (beside the east wall), while the strip between B5 and the garden's low wall grows to 0.49 m; (4) the east soakaway comes 0.13 m nearer the east wall (2.02 m).

If the setback is measured from the wall (the field value at B2): the margin at B2 is **0.40 m whatever the option** (B2 does not move) and, with the wall 0.68 m nearer along the whole cut wall, the notch gives 0.24 m as set out and 0.18 m turned: **the rotation is not what makes the cut wall tight; the wall's real position is.** So: **do the notch check first (sheet row 18); if it reads ≥ 2.46 m, turn the blocks by 0.30 m; if less, hold at 0.20 m or wait for the official survey.** The decision stands: no reduction of the cut wall until the survey. The 0.40 m option is also inside every rule against the deed lines (tightest 0.70 m) but costs B1 0.17 m and the notch 0.08 m for 0.10 m more at the south; 0.30 m is the better balance.

Set-out table of the recommended option (outer wall faces) · the sides and the diagonals unchanged

Corner	x east	y along M	Tape 1 (M1)	Tape 2 (D)	Tape 3 (R)	Tape 4 (S)	The block moves on site
B1	0.79	5.06	M1 → **5.13**	D → **23.67**	R → **30.05**	S → **33.24**	0 east, +0.13 south (0.13 m)
B2	11.04	4.94	M1 → **12.09**	D → **28.28**	R → **32.50**	S → **30.00**	**unchanged, keep the block**
B3	14.42	10.99	M1 → **18.13**	D → **26.12**	R → **28.68**	S → **23.56**	+0.07 east, −0.04 south (0.09 m)
B4	14.64	29.15	M1 → **32.62**	D → **20.17**	R → **16.90**	S → **5.58**	+0.30 east, −0.04 south (0.30 m)
B5	1.09	29.31	M1 → **29.33**	D → **6.73**	R → **6.17**	S → **16.04**	+0.30 east, +0.12 south (0.32 m)

Side / diagonal	m (unchanged)
B1–B2	**10.25**
B2–B3	**6.93**
B3–B4	**18.16**
B4–B5	**13.56**
B5–B1	**24.25**
B1–B4	**27.78**
B2–B5	**26.33**
B1–B3 (the cut's check)	**14.87**
Perimeter	**73.15**

On site, in this order: keep B2's block. Re-set **B1** from M1 5.13 and D 23.67 (it moves 0.13 m south along the string M1 → M4, x unchanged); check B1 → B2 = 10.25. Re-set **B3** from B2 6.93 and M1 18.13 check B1 → B3 = 14.87. Re-set **B4** from B3 18.16 and M1 32.62 check B1 → B4 = 27.78. Re-set **B5** from B4 13.56 and M1 29.33 check B2 → B5 = 26.33 and B1 → B5 = 24.25. Then the R / S tapes on every corner. The moves are 4–30 cm: an arc crossing is too vague for them, so measure each move from the old block along the direction given and confirm with the diagonals. Before any block moves: the notch check (row 18) and the owner's decision; then `design_gf.typ` gets the rotation and the set-out sheet is re-issued (not done here).

Rotation about B2 study · Study checks passed: B2 fixed in every option (max move 0.00 m) · B5 moves east by exactly the option (error ≤ 1 mm) · the sides and the diagonals unchanged (≤ 1 mm) in every option: a pure rotation · the house unchanged in the design files (shift 0.91, 5.06 the outline asserted): study only · the field calibration: the lane wall 0.68 m nearer than the deed line at B2 (0 … 1 m plausible), B1 → the yard wall 4.77 m = the field value · every option keeps ≥ 0.50 m to the deed setback lines (tightest 0.70 m) and every corner inside the envelope · the recommended option (0.30 m): palms ≥ 3 m (5.70 / 7.01), the gas cabinet ≥ 1.5 m from the calibrated lane wall (1.66), the tank ≥ 5 m from B5 (8.48), the car bay ≥ 0.4 m off the north wall (1.61), the swept path clear (1.96) · the route at B1 unchanged by every option (B1 moves only along the M line: ≤ 2 mm east). Field values: the owner on site, day 1 (4 Nov 2026), recorded in docs/SITE_SETOUT.md § 7. The model's house, B1 … B5, is untouched: the design files and the set-out values change only after the owner's decision.